

End-to-End Tests – Transport Module

1. Overview

This document presents end-to-end test scenarios for the Transport Management Module. The tests simulate full system behavior by running the Main program and verifying that the system correctly handles valid and invalid delivery scenarios.

The goal is to ensure that the system works correctly as a whole, including:

- Data initialization
- Business logic validation
- Delivery creation
- Output generation

2. Test Environment

- Language: Java
- Execution method: Running Main class
- Command:

```
java Main
```

The system uses in-memory data initialized inside the Main class.

3. Test Scenarios

3.1 Scenario 1 – Invalid Delivery (Incompatible Driver)

Description:

Attempt to create a delivery where the driver does not have the required license for the truck.

Input Conditions:

- Truck requires license type: C
- Driver has license type: B
- Valid weight and documents

Expected Result:

- Delivery creation should fail
- Delivery should NOT be added to the system

Actual Output:

Invalid delivery creation failed as expected.

Result: Passed

3.2 Scenario 2 – Valid Delivery Creation

Description:

Create a delivery with valid driver, truck, documents, and weight.

Input Conditions:

- Driver license matches truck requirements
- Weight is within allowed capacity ($14000 \leq 15000$)
- Valid delivery documents

Expected Result:

- Delivery should be created successfully
- Status should be set to READY
- Delivery should be stored in the system

Actual Output:

Delivery created successfully.

Result: Passed

3.3 Scenario 3 – System Output Verification

Description:

Verify that the system prints correct delivery details after execution.

Expected Output Includes:

- Delivery ID
- Date and time
- Source site
- Driver and truck details
- Weight and status
- Destinations and transported items

Actual Output:

All deliveries in the system:

Delivery ID: 1

Date: 2026-04-15

Departure Time: 08:30

Source: Tnuva Supplier

Driver: David Levi
Truck: 123-45-678
Recorded Weight: 14000.0
Status: READY
Destinations:
Document ID: 1001
Destination: Beer Sheva Branch
Items:
- Milk 3% | Quantity: 100
- Bread | Quantity: 50
Document ID: 1002
Destination: Dimona Branch
Items:
- Cheese | Quantity: 30

Result: Passed

4. Additional Validation (Unit Tests Summary)

In addition to end-to-end tests, the system includes unit tests that validate core logic such as:

- Driver–truck compatibility
- Weight validation
- Delivery creation rules
- Repository operations

All tests passed successfully:

- Total tests: 10
- Passed: 10
- Failed: 0

5. Conclusion

The Transport Management Module was tested successfully using end-to-end scenarios.

The system correctly:

- Prevents invalid delivery creation
- Allows valid delivery creation
- Maintains correct delivery state
- Produces accurate and structured output

All major system flows behave as expected.

